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Characterization of an Atmospheric Argon Plasma via Collisional-Radiative Modeling

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Abstract

Despite the wide application of Atmospheric Pressure Plasmas (APPs) in industry and science, accurate measurement of their plasma parameters remains challenging. Thus, a Collisional-Radiative Model (CRM) for argon was implemented to derive the electron temperature using optical emission spectra. The implementation was based on existing formulations in the literature. Building upon these, a highly modular, open-source Python package was developed to establish a strong foundational framework. Additionally, a spectral analysis tool was designed to calculate the relative population densities of excited argon states from the emission profile. The CRM exhibits strong agreement with a model from recent literature, although some discrepancies should be resolved in the future. Application of the model to an APP spectrum yielded inconclusive results due to hardware limitations during experimental spectral acquisition. Nevertheless, subsequent application to a low-pressure plasma demonstrated the high flexibility and accuracy of the system. Ultimately, the developed framework provides a highly adaptable, open-source foundation for spectroscopic diagnostics of argon plasmas.

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Acronyms

APP Atmospheric Pressure Plasma

BP Boltzmann Plot

BT-CCD Back-Thinned Charged-Coupled Device

CRM Collisional Radiative Model

DBD Dielectric Barrier Discharge

EEDF Electron Energy Distribution Function

EHD Electro Hydro Dynamic

FWHM Full Width Half Maximum

KIT Kyoto Institute of Technology

LPP Low-Pressure Plasma

LTE Local Thermal Equilibrium

OES Optical Emission Spectroscopy

1 Introduction

Atmospheric Pressure Plasmas (APPs) are used in a wide range of industrial and biomedical applications. They are characterized by high operating pressures, a short mean free path for electrons, and can exist in both Local Thermal Equilibrium (LTE) and non-LTE states. In fact, arc discharges can supply sufficient energy within the core to create LTE conditions at atmospheric pressures. They are primarily used for heating gases, as well as welding and cutting metals. Conversely, non-thermal *glow discharges* are less energetic; they mainly heat the electrons while keeping the heavy particles and the operating gas close to room temperature. Notable industrial applications include the cleaning of factory exhaust gases, surface treatments (decontamination, etching, coating, etc.), and gas synthesis [1]. Moreover, they are used in biomedicine for the inactivation of microorganisms, sterilization, oncology, dentistry, and the activation of metabolic functions in plants [2].

To generate these non-thermal plasmas efficiently and stably at atmospheric pressure, various discharge configurations can be employed. Among these, Dielectric Barrier Discharges (DBDs) are particularly prominent. They were first introduced by Siemens [3] in 1857 as a method to generate ozone. Since then, they have been used in a variety of technologies, including flat television screens, lasers, and surface treatments [4]. The basis of this technology is a comparatively simple geometry consisting of two electrodes separated by one or more dielectric barriers. The electrodes are driven by a high-voltage alternating current, typically in the 500 Hz to 500 kHz range. At sufficiently high voltages, an electrical breakdown occurs between the electrodes. However, the dielectric layer prevents it from developing into a full arc discharge. Instead, numerous independent micro-discharges form within the gap. These channels are weakly ionized and collectively produce a macroscopically uniform discharge behavior. The accumulation of charge on the surface of the dielectric progressively weakens the discharges, thereby limiting the total discharge current and power consumption [4].

Currently, a project at the Kyoto Institute of Technology (KIT) is investigating the activation of phytoplankton reproduction by directing an APP jet at water containing these organisms. The plasma is generated using a DBD in a cylindrical geometry. Argon gas flows axially through the electrodes and exits at the bottom, above the water vessel. Following the treatment, the mitochondria within the phytoplankton cells are marked with a fluorescent dye and examined under a microscope. Studies indicate that the production of OH radicals, induced by the plasma interaction, positively impacts the reproductive capabilities of phytoplankton [5]. Furthermore, a spatial Electro Hydro Dynamic (EHD) simulation is being developed to model the plasma dynamics and determine key plasma parameters. As a next step, the electron temperature will be determined experimentally and compared with the simulation to verify the theoretical results.

Measuring the plasma parameters of APPs is challenging because most traditional methods fail for the given high-density environment. Specifically, a Langmuir probe analysis is infeasible because the mean free path of the electrons is much smaller than the characteristic length of the probe. Accordingly, the collisionless criterion within the plasma sheath is not fulfilled, ultimately preventing the application of the sheath theory. Alternatively, Optical Emission Spectroscopy (OES) measurements are possible, but the type of analysis depends on the thermodynamic conditions of the given plasma. For plasmas in LTE, the number densities of the excited atomic states are dominated by collisional excitations. Thus, a Boltzmann distribution can be assumed for the populations of the excited atomic states and the electron temperature can be calculated from a simple linear fit to the reduced number density. However, in non-LTE plasmas, such as the DBD plasma considered in this report, the population densities are dictated by the interplay of all possible excitation and de-excitation processes instead of only the thermodynamic state. Crucially, APPs are heavily influenced by de-excitations through collisions with ground-state atoms, which further prevents the application of a corona equilibrium assumption, where radiative decay is the dominant de-excitation process. Therefore, the most promising approach is to implement a Collisional Radiative Model (CRM) of the atomic states of the argon atom, incorporating all relevant transitions. This model can be combined with OES measurements to determine the electron temperature, and, in setups with an absolute energy flux calibration, the electron density as well.

The primary objective of this project is to determine the electron temperature of the APP used in the existing phytoplankton activation setup. To achieve this, a CRM is developed from scratch within an object-oriented PYTHON environment. This model yields a quasi-steady-state solution for the population densities of the excited argon states. Parallel to modeling, an optical emission spectrum of the DBD plasma is acquired and analyzed to extract the relative population densities corresponding to the visible emission lines. Ultimately, the electron temperature is derived by coupling this experimental data with the CRM. Specifically, the experimentally obtained population density ratios are fed into an optimization algorithm, which iteratively adjusts the electron temperature in the CRM until the simulated ratios converge with the measured values.

The subsequent chapter details the implementation of the CRM and presents a parameter variation study to validate the model. Next, the experimental setup is introduced, together with a comprehensive description of the spectroscopic measurements and data analysis. Following this, the methodology for combining the experimental data with the CRM to compute the electron temperature is outlined in detail. Finally, the results are summarized and critically discussed, concluding with an outlook on future research directions.

2 Collisional-Radiative Model

Depending on the properties of interest, there are numerous approaches to model plasmas. Particle-in-cell or fluid models give the spatial distribution of plasma properties. Global plasma models aim to capture the state of different particle species and their energy distributions. CRMs focus on the atomic state of the constituent gas by simulating all possible transitions between different states of the electron shell. Precise knowledge of the number densities of all excited states is particularly important to simulate the optical emission spectrum of a plasma. This is because the intensity of an emission line depends on the number density of the upper energy level of the transition and the corresponding transition probability. By comparing the inferred population densities from an OES measurement with the population densities from a CRM, important plasma properties such as electron temperature and electron density can be determined.

2.1 Mathematical Formulation

A CRM for argon was implemented mainly based on the work of Vlcek [6], while also utilizing more recent implementations of the same model from Akatsuka [7] and Bogaerts et al. [8] as methodological support. First, the electronic structure of the argon atom is divided into 65 lumped sublevels of approximately equal excitation energies, as given in Tab. 2.1. The configurations are presented in Racah notation, because the energy levels for the noble gas argon are dominated by J_c-K-coupling. Unprimed orbitals correspond to the lower energy core state with $J = 3/2$, while primed ones refer to a core with $J = 1/2$. Most of the lower energy levels consist of single configurations. Higher levels are consistently lumped together into larger groups. Finally, the highly excited Rydberg states are only sorted by their principal quantum number and the core configuration.

The following types of transitions are included in this model. Note that the index notation follows Vlcek's notation, where the first index represents the destination and the second the source of the transition [6]. This makes the transition to matrix elements for the implementation more intuitive:

(1) Electron impact excitation and de-excitation

$\mathrm{Ar}(m) + \mathrm{e} \underset{F_{mn}}{\overset{C_{nm}}{\rightleftharpoons}} \mathrm{Ar}(n) + \mathrm{e}$

Table 2.1: Data characterising the excited levels considered in the model and the transition-dependent parameters relating to the cross sections for excitation by electrons from the ground state. A: optically allowed; P: parity-forbidden; S: spin-forbidden transitions. Taken from [6].

Level number n	Designation $n_{pqn}l[K]_J$	Excitation energy ε_{1n} / eV	Statistical weight g_n
1	$3p^6$	0.000	1
2	$4s[3/2]_2$	11.548	5
3	$4s[3/2]_1$	11.624	3
4	$4s'[1/2]_0$	11.723	1
5	$4s'[1/2]_1$	11.828	3
6	$4p[1/2]_1$	12.907	3
7	$4p[3/2]_{1,2}$, $[5/2]_{2,3}$	13.116	20
8	$4p'[3/2]_{1,2}$	13.295	8
9	$4p'[1/2]_1$	13.328	3
10	$4p[1/2]_0$	13.273	1
11	$4p'[1/2]_0$	13.480	1
12	$3d[1/2]_{0,1}$, $[3/2]_2$	13.884	9
13	$3d[7/2]_{3,4}$	13.994	16
14	$3d'[3/2]_2$, $[5/2]_{2,3}$	14.229	17
15	$5s'$	14.252	4
16	$3d[3/2]_1$, $[5/2]_{2,3} + 5s$	14.090	23
17	$3d'[3/2]_1$	14.304	3
18	5p	14.509	24
19	$5p'$	14.690	12
20	4d + 6s	14.792	48
21	$4d' + 6s'$	14.976	24
22	$4f'$	15.083	28
23	4f	14.906	56
24	$6p'$	15.205	12
25	6p	15.028	24
26	$5d' + 7s'$	15.324	24
27	5d + 7s	15.153	48
28	$5f'$, g'	15.393	64
29	5f, g	15.215	128
30	$7p'$	15.461	12
31	7p	15.282	24
32	$6d' + 8s'$	15.520	24

Table 2.1: (Continued)

Level number n	Designation $n_{pqn}l[K]_J$	Excitation energy ε_{1n} / eV	Statistical weight g_n
33	6d + 8s	15.347	48
34	6f′, g′, h′	15.560	108
35	6f, g, h	15.382	216
36	8p′	15.600	12
37	8p	15.423	24
38	7d′ + 9s′	15.636	24
39	7d + 9s	15.460	48
40	7f′, g′, h′, i′	15.659	160
41	7f, g, h, i	15.482	320
42	8d′, f′, ...	15.725	240
43	8d, f, ...	15.548	480
44	9p′, d′, f′, ...	15.769	320
45	9p, d, f, ...	15.592	640
46	$n_{pqn} = 10'$	15.801	400
47	$n_{pqn} = 10$	15.624	800
48	$n_{pqn} = 11'$	15.825	484
49	$n_{pqn} = 11$	15.648	968
50	$n_{pqn} = 12'$	15.843	576
51	$n_{pqn} = 12$	15.666	1152
52	$n_{pqn} = 13'$	15.857	676
53	$n_{pqn} = 13$	15.680	1352
54	$n_{pqn} = 14'$	15.868	784
55	$n_{pqn} = 14$	15.691	1568
56	$n_{pqn} = 15'$	15.877	900
57	$n_{pqn} = 15$	15.700	1800
58	$n_{pqn} = 16'$	15.884	1024
59	$n_{pqn} = 16$	15.707	2048
60	$n_{pqn} = 17'$	15.890	1156
61	$n_{pqn} = 17$	15.713	2312
62	$n_{pqn} = 18'$	15.895	1296
63	$n_{pqn} = 18$	15.718	2592
64	$n_{pqn} = 19'$	15.899	1444
65	$n_{pqn} = 19$	15.722	2888

(2) Electron impact ionization and three body recombination

$$\mathrm{Ar}(m) + \mathrm{e} \underset{O_m}{\overset{S_m}{\rightleftharpoons}} \mathrm{Ar}^+ + \mathrm{e} + \mathrm{e}$$

(3) Atom impact excitation and de-excitation

$$\mathrm{Ar}(m) + \mathrm{Ar}(1) \underset{L_{mn}}{\overset{K_{nm}}{\rightleftharpoons}} \mathrm{Ar}(n) + \mathrm{Ar}(1)$$

(4) Atom impact ionization and three body recombination

$$\mathrm{Ar}(m) + \mathrm{Ar}(1) \underset{W_m}{\overset{V_m}{\rightleftharpoons}} \mathrm{Ar}^+ + \mathrm{e} + \mathrm{Ar}(1)$$

(5) Radiative decay and photoexcitation

$$\mathrm{Ar}(m) + h\nu_{mn} \underset{A_{mn}}{\overset{(1-\Lambda_{nm})A_{nm}}{\rightleftharpoons}} \mathrm{Ar}(n)$$

(6) Radiative recombination and photoionization

$$\mathrm{Ar}(m) + h\nu \underset{R_m}{\overset{(1-\Lambda_m)R_m}{\rightleftharpoons}} \mathrm{Ar}^+ + \mathrm{e}$$

(7) Metastable quenching and collisional de-excitation for levels 2 and 4

Using the defined rate coefficients for the given processes, the time evolution of the number density of level n_m can be defined. Processes that increase the number density are excitations from lower levels, de-excitations from higher levels, and the three recombination terms originating from the continuum population (ions). At the same time, the number density is decreased by excitations and de-excitations outgoing from level m, as well as ionization processes (and diffusion/collisional recombination for the metastable 4s states). Thus, the population rate can be expressed as:

$$\begin{aligned}
\frac{\mathrm{d}n_m}{\mathrm{d}t} = &+ n_n \left[\sum_{n<m} (C_{mn}n_e + K_{mn}n_1) + \sum_{n>m} (F_{mn}n_e + L_{mn}n_1 + \Lambda_{mn}A_{mn}) \right] \\
&+ n_e n_+ (O_m n_e + \Lambda_m R_m + W_m n_1) \\
&- n_m \Bigg[\sum_{n>m} (C_{nm}n_e + K_{nm}n_1) + \sum_{n<m} (F_{nm}n_e + L_{nm}n_1 + \Lambda_{nm}A_{nm}) \\
&\qquad + S_m n_e + V_m n_1 + \frac{D_m}{\Lambda^2} + n_1^2 B_m \Bigg],
\end{aligned} \tag{2.1}$$

where Λ_{mn} and Λ_m are optical escape factors for inter-state and continuum transitions respectively, D_m is the diffusion coefficient, $\Lambda = L/\pi$ is the characteristic diffusion length, and B_m is the rate constant for depopulation by three-body collisions with ground state atoms. Both constants are non-zero only for the two metastable 4s states (levels 2 and 4) [7].

Here, it is assumed that the fundamental diffusion mode is dominant in the cylindrical discharge tube (only diffusion in the radial direction) and that the discharge tube is much longer than its radius, so that the characteristic length can be approximated as $\Lambda = 2.405/R$. Furthermore, the temperature dependence of the diffusion constant is assumed to be the same as for neon, such that $D_n(T_a) = D_n(T_{\mathrm{ref}}) \cdot (T_a/T_{\mathrm{ref}})^{0.73}$. Using these relations, the diffusion term can be written as $D_n/\Lambda^2 = d_n T_a^{0.73} n_1 R^2$, where $d_n = D_n \cdot 2.405^2/T_{\mathrm{ref}}^{0.73}$ [6]. The numerical values for D_n and B_n are taken from Tachibana [9], who determined them using a drift-tube technique.

The rate equation given in Eq. 2.1 can be simplified to a set of coupled linear equations of the form

$$\sum_{n=2}^{65} a_{mn} n_n = -\delta_m - a_{m1} n_1, \tag{2.2}$$

when assuming a quasi-stationary state, where $\mathrm{d}n_m/\mathrm{d}t = 0$ for all populations. For this, the ground state transitions are extracted from the sums and transposed, which is why they appear next to the recombination source term. By comparing the two sums over the excitation and de-excitation rates in Eq. 2.1, it becomes clear that the two terms are identical when exchanging n and m. Therefore, the sum in the loss term can be simplified to the sum over all matrix elements for transitions outgoing from level m. The following terms define all elements of Eq. 2.2:

$$a_{mn} = n_e C_{mn} + n_1 K_{mn} \qquad m > n \tag{2.3}$$

$$a_{mn} = n_e F_{mn} + n_1 L_{mn} + \Lambda_{mn} A_{mn} \qquad m < n \tag{2.4}$$

$$a_{mm} = -\Big(n_e S_m + n_1 V_m + \sum_{\substack{n=1 \\ n \neq m}}^{65} a_{nm} + D_m/\Lambda^2 + n_1^2 B_m\Big) \qquad m = n \tag{2.5}$$

$$\delta_m = n_e n_+ (n_e O_m + n_1 W_m + \Lambda_m R_m). \tag{2.6}$$

For the rate coefficients, it is assumed that both the thermal argon atoms and the free electrons follow a Maxwellian Electron Energy Distribution Function (EEDF) denoted by $f(u)$. Here, u is the dimensionless energy $u = \varepsilon/k_B T_e$, with the electron energy ε, Boltzmann constant k_B, and electron temperature T_e. The EEDF is normalized, so that

$$2\pi \left(\frac{2k_B T_e}{m_e}\right)^{3/2} \int_0^\infty \sqrt{u} f(u) du = 1. \tag{2.7}$$

The rate coefficients for electron impact excitation/de-excitation (C_{mn}/F_{mn}) and ionization/recombination (S_n/O_m) are [10]:

$$C_{mn} = 8\pi \left(\frac{k_B T_e}{m_e}\right)^2 \int_{u_{mn}}^{\infty} f(u)\sigma_{nm}(u)u\,\mathrm{d}u \tag{2.8}$$

$$F_{mn} = 8\pi \left(\frac{k_B T_e}{m_e}\right)^2 \frac{g_m}{g_n} \int_{u_{mn}}^{\infty} f(u-u_{mn})\sigma_{mn}(u)u\,\mathrm{d}u \tag{2.9}$$

$$S_n = 8\pi \left(\frac{k_B T_e}{m_e}\right)^2 \int_{u_n}^{\infty} f(u)\sigma_n(u)u\,\mathrm{d}u \tag{2.10}$$

$$O_m = 8\pi \left(\frac{k_B T_e}{m_e}\right)^2 \frac{g_m}{2g_+} \left(\frac{h^2}{2\pi m_e k_B T_e}\right)^{3/2} \cdot \int_{u_m}^{\infty} f(u-u_m)\sigma_m(u)u\,\mathrm{d}u, \tag{2.11}$$

where m_e is the electron mass, h Planck's constant, g_n the statistical weight of level n, g_+ the statistical weight of the ionic ground state, σ_{nm} the cross section for electron excitation from level n to m, σ_{mn} the reverse process, and σ_n the cross section for electron impact ionization. The formulas used for the cross sections will be defined in detail in 2.1.1.

Next, the formulas for atom impact excitation/de-excitation (K_{mn}/L_{mn}) and ionization/recombination (V_n/W_m) are adapted from Bacri and Gomes [11] and defined as:

$$K_{mn} = 2\left(\frac{2k_B T_a}{\pi\mu}\right)^{1/2} b_{nm}(\varepsilon_{mn} + 2k_B T_a)\exp\left(-\frac{\varepsilon_{mn}}{k_B T_a}\right) \tag{2.12}$$

$$L_{mn} = K_{nm}\left(\frac{g_m}{g_n}\right)\exp\left(\frac{\varepsilon_{mn}}{k_B T_a}\right) \tag{2.13}$$

$$V_n = 2\left(\frac{2k_B T_a}{\pi\mu}\right)^{1/2} b_n(\varepsilon_n + 2k_B T_a)\exp\left(-\frac{\varepsilon_n}{k_B T_a}\right) \tag{2.14}$$

$$W_m = V_m \frac{g_m}{2g_+}\left(\frac{h^2}{2\pi m_e k_B T_e}\right)^{3/2} \exp\left(\frac{\varepsilon_m}{k_B T_a}\right). \tag{2.15}$$

Here, T_a is the gas temperature, $\mu = m_a/2$ is the reduced mass of two argon atoms with mass m_a, ε_{mn} the excitation energy between levels n and m, ε_n the ionization energy of level n, and b_{nm} and b_n are energy-dependent semi-empirical parameters that will be specified in section 2.1.2.

Finally, the recombination rate coefficient is calculated using the cross section for photoionization σ_m^{P} that will be elaborated on in section 2.1.3. The initial formula given by Vlcek and Pelikan [10] was adjusted so that it is explicitly expressed as a function of u by substituting the photon frequency:

$$R_m = 8\pi \frac{k_B T_e}{m_e^3 c^2}\left(\frac{g_m}{2g_+}\right)\int_0^{\infty} f(u)\sigma_m^{\mathrm{P}}(u+u_m)(u k_B T_e + \varepsilon_m)^2\,\mathrm{d}u, \tag{2.16}$$

where c is the speed of light in vacuum, u_m is the dimensionless ionization energy of level m, and the integral is evaluated over the dimensionless energy of the electron.

Additionally, some given rate coefficient expressions need to be multiplied by additional factors for certain transitions. Specifically, the excitation and ionization coefficients for atom-atom collisions (K_{mn}/V_n), must be multiplied by a factor of 0.5 for transitions from the lower level $n = 1$, because the involved atoms are indistinguishable. Moreover, both de-excitation coefficients for electronic and atomic collisions (C_{mn}/K_{mn}) that lead to the ground state $m = 1$ must use $g_m = 2/3$ and $g_m = 1/3$ for transitions from a core with $J_c = 3/2$ and $J_c = 1/2$, respectively. Likewise, the statistical weight for the ionic ground state g_+ used for the recombination coefficients ($R_m/O_m/W_m$) must be set to 4 or 2 for an unprimed or primed core, respectively. Lastly, the ionic ground state density n_+ used in Eq. 2.6 must also be set to $2/3 \cdot n_e$ or $1/3 \cdot n_e$ for an unprimed or primed core, respectively.

Since the mathematical foundation of the CRM results in a linear matrix equation, finding a solution is straightforward with modern computational tools. The primary challenge lies in describing the cross sections accurately, even for transitions where little to no experimental data is available. Thus, the following sections describe the definition of the cross sections for all processes in detail and explain the sources of all empirical parameters used.

2.1.1 Cross Sections for Electron-Atom Collisions

Initially, all electron-atom collisions are classified into three different categories of transitions. This is important because every transition type follows different analytical dependencies, and the different cross sections vary vastly in magnitude. First, the transitions with the highest probability are optically ALLOWED transitions. Such a transition is defined by the following selection rules: $\Delta l = \pm 1$ and $\Delta J = 0, \pm 1$, but not $J = 0 \to J = 0$. Next, PARITY FORBIDDEN transitions are characterized by violating the dipole selection rule so that $\Delta l \neq \pm 1$. Lastly, SPIN FORBIDDEN transitions are treated, i.e., transitions for which the total angular momentum of the core changes. These transitions are suppressed because the electron has to reach deep into the electron cloud of the atom to change the spin quantum number of the core. Thus, intercombination transitions are only considered when they follow the selection rules for allowed transitions [8].

For most of the excitation transitions, the semi-empirical formulas presented by Drawin [12] are used. They are separated into the three different transition types, where σ_{mn}^{A} marks allowed, σ_{mn}^{P} parity forbidden, and σ_{mn}^{S} spin forbidden transitions. Here, they are used as follows:

$$\sigma^A_{mn} = 4\pi a_0^2 \left(\frac{\varepsilon_1^H}{\varepsilon_{mn}}\right) f_{mn}\alpha^A_{mn} U_{mn}^{-2}(U_{mn}-1)\ln(1.25\,\beta_{mn}U_{mn}) \tag{2.17}$$

$$\sigma^P_{mn} = 4\pi a_0^2 \alpha^P_{mn} U_{mn}^{-1}(1-U_{mn}^{-1}) \tag{2.18}$$

$$\sigma^S_{mn} = 4\pi a_0^2 \alpha^S_{mn} U_{mn}^{-3}(1-U_{mn}^{-2}) \tag{2.19}$$

where a_0 is the first Bohr radius, ε_1^H is the ionization energy for atomic hydrogen from the ground state, ε_{mn} is the excitation energy between level n and m, f_{mn} is the oscillator strength for the given transition, α_{mn} and β_{mn} are the Drawin parameters for the respective transitions that are determined by comparison with experimental data, and $U_{mn} = \varepsilon/\varepsilon_{mn}$ is the normalized energy of the electron. In the following, the product of oscillator strength and scaling parameter used in Eq. 2.17 will be referred to as $\alpha'^A_{mn} = f_{mn}\alpha^A_{mn}$.

Since some energy levels (see Tab. 2.1) are lumped together, it is possible that there exist transitions for which multiple transition types are allowed. In this case, the sum of all applicable cross sections is computed and used to calculate the excitation/de-excitation rate coefficient C_{mn}/F_{mn}.

For all transitions from the ground state, Vlcek determined the respective Drawin parameters for the dominant transitions and tabulated them together with the transition types. The detailed description on how the author determined these parameters can be found in [6]. As an exception, the additive method mentioned in the previous paragraph is not used for the ground state transitions. Transitions with very low transition probabilities are neglected, and no parameters are specified.

Regarding transitions between excited levels, no experimental data is available, requiring the use of theoretically calculated cross sections. Here, data from Kimura et al. [13] is used that covers the scaling parameters for all allowed and parity forbidden multiplet transitions ($N \to M$) in the range $2 \leqslant n_{pqn} \leqslant 10$. Utilizing this data, the inter-state transitions for the levels $2 \leqslant m < n \leqslant 45$ and $2 \leqslant m < n \leqslant 47$ are calculated for allowed and parity forbidden transitions, respectively. The authors used a J_c-K-coupling scheme to calculate the transition probabilities, where they only include transitions within two core states, without taking intercombination transitions into account. In reality, the coupling for argon follows a mixture of L-S and J_c-K-coupling, allowing for intercombination transitions [6]. Thus, allowed transitions require proper handling of the given data to respect significant probabilities, especially for the lower energy levels. Parity forbidden transitions, however, are already suppressed so much that intercombinations can be neglected for the majority of the transitions [8].

The following strategy was chosen to extract the correct scaling parameters for the allowed transitions: Kimura et al. calculated both radiative transition probabilities (A_{MN}) and cross-section scaling parameters (K_{MN}) for allowed and parity forbidden transitions ($\beta_{\mathrm{MN}} = 1.00$). For calculating the parameters of the allowed transitions, they deduced the oscillator strength from their own radiative transition probabilities. Accordingly, the values $\alpha'^{A}_{\mathrm{MN}} = f_{\mathrm{MN}} \cdot K_{\mathrm{MN}}$ were derived inversely using their own A_{MN} and the formula

$$f_{\mathrm{MN}} = \frac{\varepsilon_0 m_e c^3 h^2}{2\pi e^4} \frac{g_M}{g_N} \frac{A_{\mathrm{MN}}}{E^2_{\mathrm{MN}}} \approx 2.3046 \cdot 10^{-8}\,\mathrm{eV}^2\,\mathrm{s}\, \frac{g_M}{g_N} \frac{A_{\mathrm{MN}}}{E^2_{\mathrm{MN}}}, \tag{2.20}$$

for the oscillator strength, where ε_0 is the vacuum permittivity, and E_{MN} the transition energy between the multiplets in eV [14]. Next, for each multiplet transition, the separate parameters for the two core states are averaged and redistributed among all corresponding fine-structure transitions for which an oscillator strength is tabulated in the NIST Atomic Spectra Database [15]. The weights are calculated according to $\alpha'^{A}_{mn} = \alpha'^{A}_{\mathrm{MN}} \cdot f_{mn}/\Sigma f_{mn}$. After this, the fine-structure transitions are redistributed among the 65 lumped levels defined above. For all multiplet transitions that do not have a corresponding oscillator strength listed in the NIST data, the parameters are distributed among the lumped transitions using the statistical weights of the upper level: $\alpha'^{A}_{mn} = \alpha_{\mathrm{MN}} \cdot g_n/\Sigma g_n$. This is to conserve the total cross section for the case where the upper state of the multiplet is divided into multiple levels in the CRM. Every single upper level then only receives the given percentage of the total cross section.

Concerning the highly excited Rydberg state transitions ($45 \leqslant m < n \leqslant 65$), no experimental or theoretical data is available. Instead, since the valence electron is so far away from the nucleus and the inner orbitals, the transitions are assumed to be hydrogenic. This means the scaling parameters are set to unity so that $\alpha'^{A}_{mn} = f_{mn}$ and β_{mn}=1.00. For calculating the oscillator strength, an approximate formula for principal quantum numbers n_{pqn} in hydrogenic atoms is used [16, p. 269]:

$$g_n f_{n_{pqn} \to m_{pqn}} = \frac{2^6}{3\sqrt{3}\pi} \left(\frac{1}{n^2_{pqn}} - \frac{1}{m^2_{pqn}} \right)^{-3} \frac{1}{n^3_{pqn} m^3_{pqn}}. \tag{2.21}$$

This formula calculates the weighted oscillator strength between two electron levels. To get the absorption oscillator strength, this equation must be divided by the degeneracy of the lower level $g_n = 2n^2_{pqn}$. After simplifying the fraction, the term can be simplified to

$$f_{n_{pqn} \to m_{pqn}} = \frac{32}{3\sqrt{3}\pi} \frac{n_{pqn} \cdot m^3_{pqn}}{(m^2_{pqn} - n^2_{pqn})^3}. \tag{2.22}$$

For the parity forbidden transitions, the parameters from the same data set can be directly used in Eq. 2.18. Again, the corresponding multiplet value is distributed among the levels used in this CRM by using the statistical weights of the upper levels. Furthermore, due to their low probability, changes in the core spin are disregarded for all parity forbidden transitions. Thus, the scaling parameters for the two cores are processed individually. For all transitions that are not considered in the data set ($47 \leqslant m < n \leqslant 65$), the scaling parameter α^P is set to zero, assuming that the allowed transitions are dominant. Importantly, for the parity forbidden transitions within the 4s states (levels 2-5), the semi-empirical cross section formulas calculated by Baranov et al. [17] are utilized instead of the Drawin formulas. In their original formulation for the de-excitations for the transitions $4s[3/2]_1 \rightarrow 4s[3/2]_2$ (level 3→2) and $4s'[1/2]_1 \rightarrow 4s'[1/2]_0$ (level 5→4), the equations take the form

$$\sigma_{32} = 2.2 \cdot 10^{-14} e^{-16} \varepsilon^{-0.54} \tag{2.23}$$

$$\sigma_{54} = 1.9 \cdot 10^{-14} e^{-31.4} \varepsilon^{-1.04}, \tag{2.24}$$

where e is Euler's number and ε the electron energy in erg. To retrieve the formula for the electron excitation cross section, we need to translate to the reverse process using the relation derived by Klein and Rosseland [18]. The reverse process demands that the electron cross section is evaluated at the energy it had before excitation minus the excitation energy:

$$\sigma_{23} = \frac{g_3}{g_2} \frac{\varepsilon - \varepsilon_{32}}{\varepsilon} \cdot 2.2 \cdot 10^{-14} e^{-16} (\varepsilon - \varepsilon_{32})^{-0.54} \tag{2.25}$$

$$\sigma_{45} = \frac{g_5}{g_4} \frac{\varepsilon - \varepsilon_{54}}{\varepsilon} \cdot 1.9 \cdot 10^{-14} e^{-31.4} (\varepsilon - \varepsilon_{54})^{-1.04}. \tag{2.26}$$

With some algebra, the constants are combined, together with the conversion factor from erg to eV, and the normalized energy $U_{mn} = \varepsilon/\varepsilon_{mn}$ is introduced, yielding the final expressions that replace the Drawin formulas for the given transitions

$$\sigma_{23} = 5.7965 \cdot 10^{-15} \frac{g_3}{g_2} (1 - U_{32}^{-1})^{0.46} (\varepsilon_{32} U_{32})^{-0.54} \tag{2.27}$$

$$\sigma_{45} = 8.1098 \cdot 10^{-16} \frac{g_5}{g_4} (1 - U_{54}^{-1})^{-0.04} (\varepsilon_{54} U_{54})^{-1.04}. \tag{2.28}$$

Finally, for the spin forbidden transitions, no experimental data is available. Thus, it is approximated that all intercombination transitions follow the same dependency as σ_{23} but multiplied by a factor of 0.1 [6]. This factor is chosen because experiments for neon showed that spin forbidden transitions are suppressed to approximately 10 % of the parity forbidden transitions. All intercombination transitions that are already covered by allowed transitions (because NIST listed an oscillator strength for them) are skipped. Furthermore, for the highly excited states, the spin forbidden cross sections are assumed to be negligible and therefore set to zero ($47 \leqslant m < n \leqslant 65$).

For the ionization by electron impact, another formula proposed by Drawin [12] is used in the form

$$\sigma_m = 4\pi a_0^2 \left(\frac{\varepsilon_1^H}{\varepsilon_n}\right)^2 \xi_m \alpha_m U_m^{-2}(U_m - 1)\ln(1.25\,\beta_m U_m), \tag{2.29}$$

where ξ_m is the number of energetically equivalent electrons in level m, ε_m is the corresponding ionization energy, $U_m = \varepsilon/\varepsilon_m$ is the normalized energy, and α_m and β_m are level-dependent parameters. While the ground state has six energetically identical electrons ($\xi_m = 6$), all other states only have one ($\xi_m = 1$). The scaling parameters used here were calculated by Vlcek, based on a synthesis of available experimental data. Crucially, the two core configurations have different ionization energies from the ground state. This needs to be respected while calculating the ionization energy for any given level.

2.1.2 Cross Sections for Atom-Atom Collisions

Little experimental data is available for atom-atom collision cross sections. However, Drawin and Emard [19] proposed the following semi-empirical formula for optically allowed transitions:

$$Q_{mn} = 4\pi a_0^2 \left(\frac{\varepsilon_1^{\mathrm{H}}}{\varepsilon_{mn}}\right)^2 \frac{m_{\mathrm{Ar}}}{m_{\mathrm{H}}} \xi_m^2 f_{mn} \frac{2m_e}{m_e + m_{\mathrm{Ar}}} \tag{2.30}$$

$$\cdot\,(U_{mn} - 1)\left(1 + \frac{2m_e}{m_e + m_{\mathrm{Ar}}}(U_{mn} - 1)\right)^{-2}, \tag{2.31}$$

where m_{Ar} and m_{H} are the masses of argon and hydrogen, respectively, and $U_{mn} = E/E_{mn}$ is the normalized kinetic impact energy. At energies close to the threshold, the cross sections derived from this formula exhibit an approximately linear dependence according to

$$Q_{mn} = b_{mn}(E - E_{mn}), \tag{2.32}$$

where b_{mn} is a transition-dependent scaling parameter. Following the calculations of Bacri and Gomes [11], these parameters were determined by performing a fit to the theoretical values from Eq. 2.31 and some available experimental data. Vlcek [6] determined the following relationship between the excitation energy of a transition and the scaling parameter:

$$b_{mn} = 8.69 \cdot 10^{-18} \varepsilon_{mn}^{-2.26}. \tag{2.33}$$

This relationship is used for all transitions except those between the lowest four excited states. For these, the constant $8.69 \cdot 10^{-18}$ is replaced by the value $1.79 \cdot 10^{-20}$ for transitions within a core state and to $4.80 \cdot 10^{-22}$ for intercombination transitions.

Concerning ionization through atom-atom collisions, the same linear approximation and the same scaling parameter relationship is used. Here, the excitation energy is simply replaced by the ionization energy for the respective levels.

2.1.3 Cross Sections for Photoionization

The photoionization cross sections σ_m^P are mainly based on the work of Katsonis [20], with minor modifications to the 4s and 4s′ state formulas suggested by Vlcek [6]. Utilizing experimental data and theoretical calculations, Katsonis constructs a piecewise definition of the cross sections. For the ground state formulas, sufficient data was available to allow for an empirical fit, producing the relationships

$$\sigma_1^{\mathrm{P}} = \begin{cases} 3.5 \cdot 10^{-17} & \varepsilon_1 \leqslant h\nu \leqslant 2\varepsilon_1^{\mathrm{H}} \\ 2.8 \cdot 10^{-16} \left(\frac{\varepsilon_1^{\mathrm{H}}}{h\nu}\right)^3 & h\nu > 2\varepsilon_1^{\mathrm{H}}, \end{cases} \tag{2.34}$$

where ν is the frequency of the photon. While the cross section is completely energy independent in the region close to the threshold, a subsequent $1/(h\nu)^3$ drop-off follows after crossing two times the ionization energy of hydrogen.

For excited states from the 4p orbital up to $n_{pqn} = 9$ (levels $6 \leqslant n \leqslant 45$), Katsonis uses a variation of Kramers' formula. The original formula defines the photoionization cross section for a hydrogen electron with principal quantum number n_{pqn}:

$$\sigma_m^P = \frac{7.91 \cdot 10^{-18}}{(h\nu)^3 n_{pqn}^5}. \tag{2.35}$$

Katsonis [20] introduces the effective quantum number $n_{pqn}^* = \sqrt{\varepsilon_1^{\mathrm{H}}/\varepsilon_m}$ from the ratio of the ionization energy of hydrogen and the ionization energy of level m. Moreover, he defines the normalized photon energy $h\nu/\varepsilon_1^{\mathrm{H}}$. This leads to a generalized formula, applicable to non-hydrogenic atoms, of the form

$$\sigma_m^{\mathrm{P}} = \sum_{(n_{pqn},l)} \gamma(n_{pqn}, l) \cdot 7.91 \cdot 10^{-18} \left(\frac{\varepsilon_m}{\varepsilon_1^{\mathrm{H}}}\right)^{2.5} \left(\frac{\varepsilon_1^{\mathrm{H}}}{h\nu}\right)^3, \tag{2.36}$$

where $\gamma(n_{pqn}, l)$ is a parameter that distributes the cross section for a given principal quantum number among all possible orbitals. Katsonis calculated these values based on data from Burgess [21] and tabulated them. For $n_{pqn} > 9$, the separation due to angular momentum becomes insignificant, so that the electron behaves hydrogenically with $\gamma(n_{pqn}) = 1$. However, for lower levels, the orbitals in the present model are divided further by the total angular momentum j. Thus, the tabulated $\gamma(n_{pqn}, l)$ need to be weighted according to the fine-structure. For this, Katsonis suggests

$$\mu(n_{pqn}, l, j) = \frac{g(n_{pqn}, l, j)}{4(J_c + 1/2)(2l + 1)}. \tag{2.37}$$

In the intermediate region with the 4s states (levels $2 \leqslant n \leqslant 5$), the electron is still very close to the inner electrons so that its cross section is affected by the interaction with the other orbitals. Therefore, Katsonis chose to approximate the cross sections by a combination of the ground state (with the plateau for low energies) and the generalized Kramers' formula for higher energies:

$$\sigma_m^{\mathrm{P}} = \begin{cases} 2 \cdot 10^{-18} \gamma(n_{pqn}, l) & \bar{\varepsilon}_{4s} \leqslant h\nu \leqslant 0.59 \varepsilon_1^{\mathrm{H}} \\ 7.91 \cdot 10^{-18} \gamma(n_{pqn}, l) \left(\frac{\bar{\varepsilon}_{4s}}{\varepsilon_1^{\mathrm{H}}}\right)^{2.5} \left(\frac{\varepsilon_1^{\mathrm{H}}}{h\nu}\right)^3 & h\nu > 0.59 \varepsilon_1^{\mathrm{H}}. \end{cases} \tag{2.38}$$

Here, $\bar{\varepsilon}_{4s}$ is the mean ionization energy of all 4s and 4s' states, and the parameters for level 2, 3, 4, and 5 are given as $\gamma(n_{pqn}, l) = 0.0763, 0.048, 0.0305, 0.0915$.

In short, the following procedure is used for the calculation of the photoionization cross sections: For the ground state, the semi-empirical formulas given in Eq. 2.34 are used, where no additional weighting parameters are needed. Next, the terms for the intermediate states are calculated with Eq. 2.38 and the given weighting parameters from Vlcek. Finally, for the higher excited states, the generalized Kramers' formula in Eq. 2.36 is used together with the prepared effective level-dependent weighting parameters $\gamma(n_{pqn}, l)$, where states above $n_{pqn} = 9$ are treated as hydrogenic.

2.1.4 Transition Probabilities and Escape Factors

The dataset for the transition probabilities is compiled from three different sources. For the most important and dominant transitions, the suggested values from the NIST database are used [15]. These lie in the transition energy range of $0.5\,\mathrm{eV} \leqslant \varepsilon_{mn} \leqslant 14.3\,\mathrm{eV}$ and are restricted to measurable transitions. As already mentioned above, Kimura et al. [13] used $J_c - K$-coupling to calculate the transition probabilities in the range $2 \leqslant n_{pqn} \leqslant 10$. For all multiplet transitions that are not already specified in the NIST database, these values are used after distributing them among the fine-structure analogously to the distribution for the scaling parameters outlined in section 2.1.1. Lastly, for all transitions in the range $45 \leqslant m < n \leqslant 65$, the transition probabilities are calculated using the oscillator strengths based on the hydrogenic assumption in Eq. 2.22. The two quantities are related by

$$A_{n_{pqn} \to m_{pqn}} = \frac{2\pi e^4 \varepsilon_{mn}^2}{m_e c^3 \varepsilon_0 h^2} \frac{g_m}{g_n} f_{n_{pqn} \to m_{pqn}} \tag{2.39}$$

$$= 4.3392 \cdot 10^7 \cdot \varepsilon_{mn}^2 \frac{g_m}{g_n} f_{n_{pqn} \to m_{pqn}}, \tag{2.40}$$

where ε_{mn} is the transition energy between the two levels in eV, and g_m and g_n are the respective statistical weights of the lower and upper level [22, p. 31].

High-pressure plasmas are optically thick, meaning they re-absorb a considerable amount of their own emitted radiation. This happens because for any given transition, the emitted photon can encounter an atom that currently occupies the respective lower state of the transition and re-excite it. Low-density plasmas are usually optically thin, meaning the lower state densities are simply too low for photoexcitation to occur within the plasma boundaries. However, for the high-pressure and low-temperature plasmas considered in this project, all transitions leading to the ground state will be optically thick because there is a very large ground state population available to absorb the emitted photons. Therefore, the process of radiation trapping is implemented for all optical ground state transitions through the definition of the optical escape factor. For all other transitions, the optical escape factor is set to 1.

The optical escape factor Λ_{1n} is related to the transmission coefficient T by the relation

$$\Lambda_{1n} \approx g_0 T, \tag{2.41}$$

where g_0 is a constant that depends on the absorption profile and the gap geometry. For a cylindrical geometry (as assumed in this work) $T = T(R)$ and g_0 is 1.9 for a Doppler-broadened profile and 1.3 for a profile dominated by pressure-broadening. Mills and Hieftje [23] determined the transmission coefficient for a cylindrical geometry of radius R, which is initially excited along the cylinder axis, to be

$$T(R) = T_\mathrm{D} \exp\left(\frac{-\pi T_\mathrm{CD}^2}{4T_\mathrm{C}}\right) + T_\mathrm{C}\Gamma\left(\frac{\sqrt{\pi}T_\mathrm{CD}}{2T_\mathrm{C}}\right), \tag{2.42}$$

where T_D and T_C are the transmission coefficients for pure Doppler and collisional broadening, respectively, and T_CD is the coefficient for pressure-broadened emission and Doppler-boradened absorption profiles:

$$T_\mathrm{D} = \left(k_0 R\sqrt{\pi \ln(k_0 R)}\right)^{-1} \tag{2.43}$$

$$T_\mathrm{C} = \sqrt{\frac{a}{\sqrt{\pi}k_0 R}} \tag{2.44}$$

$$T_\mathrm{CD} = \frac{2a}{\pi\sqrt{k_0 R}}, \tag{2.45}$$

where $k_0 R$ is the optical depth to the line center, and a is the damping coefficient. For these two parameters, the following relations hold for optically thick transitions [8]:

$$k_0 R = \frac{2.1 \cdot 10^{-17} g_n}{\varepsilon_n^3 \sqrt{T_a}} A_{1n} n_1 R, \tag{2.46}$$

$$a = A_{1n} \left(1 + \frac{3.225 \cdot 10^{-14}}{\varepsilon_n^3} g_n n_1 \right) \frac{4.839 \cdot 10^{-9}}{\varepsilon_n \sqrt{T_a}}, \tag{2.47}$$

2.2 Numerical Implementation

In this project, the described CRM is implemented in a class-based approach using PYTHON[1]. This language was chosen because solving a linear matrix equation for the quasi-stationary solution does not require high computational efficiency. At the same time, PYTHON offers a large variety of pre-built packages that simplify the implementation and readability of the code. The overall code architecture is schematically depicted in Fig. 2.1.

The core of the simulation lies in the *PhysicsEngine* class that builds the linear matrix equation according to Eqs. 2.2 to 2.6. For this, it needs to generate the EEDF, the cross section formulas, and calculate the rate coefficients. All these processes are outsourced to dedicated classes. After definition of the matrices, the linear equation is solved and the number densities of all levels returned.

For the end-user, all communication happens with the *Interface* class using .toml config files. One of these files is used to configure the CRM parameters by turning certain transition types on and off, or limiting the number of levels that are simulated. It is also possible to restrict the range within which certain transitions are calculated. The other file specifies the simulation parameters. This means choosing the type of simulation (simulation/optimization), specifying the plasma parameters, and adjusting the exporting options. Two different types of exports are implemented; a Boltzmann Plot (BP)[2] that is saved as a png and a raw csv data export.

When choosing the *simulation* option, the model is run for the given input parameters. Here, it is possible to perform parameter variations by handing over lists for all parameters that shall be varied. The different reduced population densities are then printed onto one shared plot to visualize trends. Moreover, the plotting options can be adjusted using a personalized *mplstyle* styling sheet.

For the *optimization* option, a set of relative experimental densities needs to be specified on top of the other parameters described for the simulation options. These densities should be determined by OES and do not need to be in absolute intensity

1 The full simulation code and the data processing pipeline are published on https://github.com/peterplr/crm-argon under the GPL-3.0 license for public use.

2 On a Boltzmann plot, the reduced number density, i.e., the number density divided by the statistical weight, is plotted for all levels. In local thermal equilibrium, the plot forms a line for which the slope is related to the electron temperature.

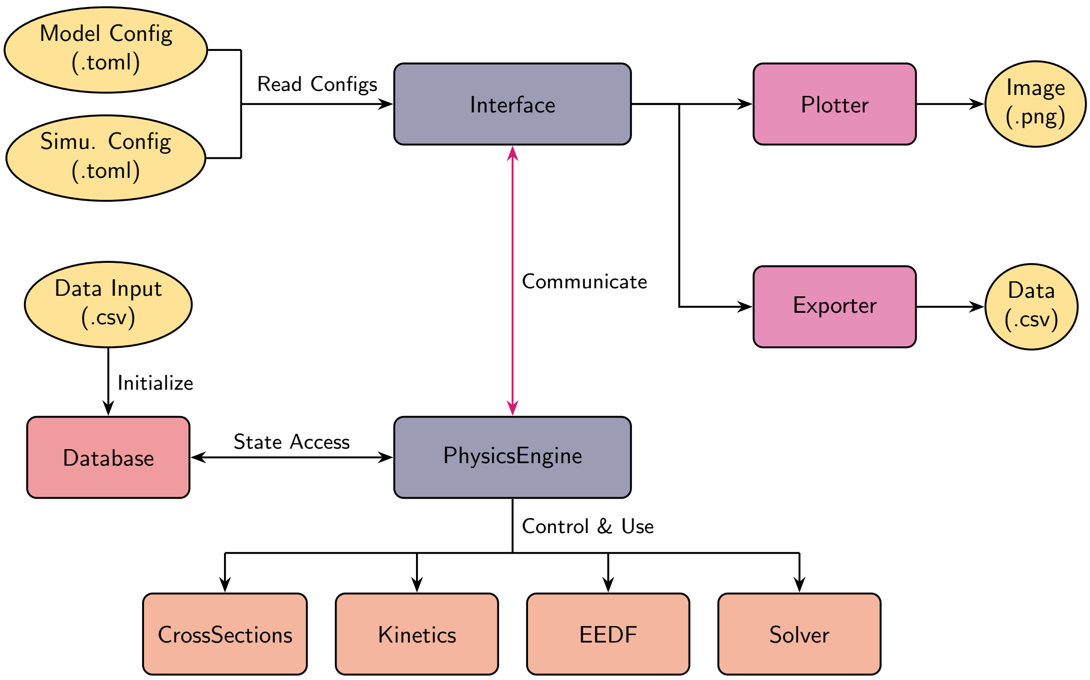

Figure 2.1: This flow chart visualizes the structure of the simulation code. The end user only communicates with the Interface via config files, while it properly handles the core simulation classes.

units. However, the spectrometer needs to be response corrected to allow for relative correlations within the spectrum. Furthermore, the densities can be weighted (w_i), i.e., to give several lines in the 4p block the same statistical weight as one line from the 5p block. Internally, the optimizer calculates all possible density ratios from the given experimental values and optimizes the parameter T_e, or, if activated, T_e and n_e. More specifically, since the number densities differ significantly in magnitude, the deviation is calculated logarithmically according to:

$$E(T_e, n_e) = \sum_{(i,j)\in P} (w_i w_j) \left[\ln\left(\frac{n_i^{\mathrm{exp}}}{n_j^{\mathrm{exp}}}\right) - \ln\left(\frac{n_i^{\mathrm{sim}}(T_e, n_e)}{n_j^{\mathrm{sim}}(T_e, n_e)}\right)\right]^2 . \tag{2.48}$$

Once the optimization converges, the optimal set of parameters is used one more time to calculate the theoretical number densities. Ultimately, a BP, showing the optimized values together with their errors, and optionally a csv of the simulation results are saved.

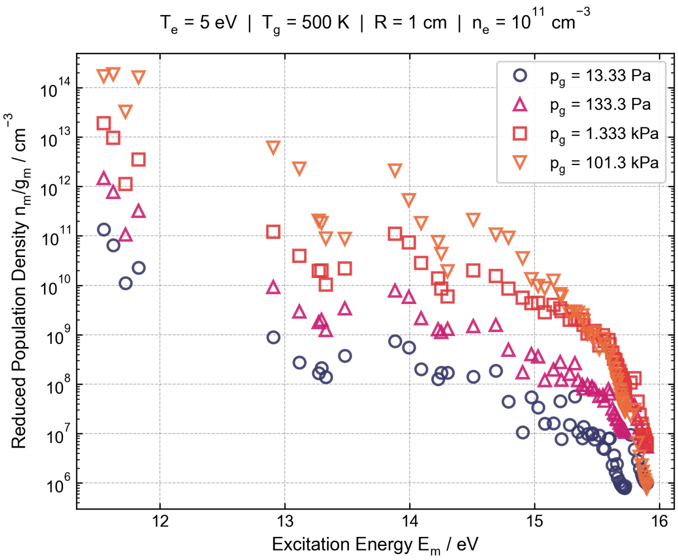

Figure 2.2: In this BP, the reduced population densities for the given input parameters are displayed for different discharge pressures. Overall, the reduced number densities rise for higher pressures, while the linear slopes within the separate energy blocks evolve towards steeper declines.

2.3 Model Testing

Before the CRM is applied to the experiment, its output is reviewed for different operating points and the results are discussed for their physical accuracy. The input parameters are chosen in accordance with Akatsuka [7], to allow for a comparison later on.

First, the discharge pressure is varied between 13.33 Pa and 101.3 kPa, for an operating point at $T_e = 5\,\text{eV}$, $n_e = 1 \cdot 10^{11}\,\text{cm}^{-3}$, $T_g = 500\,\text{K}$ and $R = 1\,\text{cm}$. The results are summarized on one single BP to visualize the different trends (see Fig. 2.2). The plot shows that the number densities of all excited levels increase for higher pressures in the discharge chamber. This is caused by the higher ground state density that increases the overall upward flux into the system. At the same time, the reduced population densities follow different linear relationships within the energy blocks of similar excitation energy. For high pressures, the number densities fall off more steeply than for low pressures. This shows how the de-excitation by collisions with ground state atoms becomes dominant for higher neutral densities. Accordingly, the model correctly visualizes the physically expected trends.

Next, the electron temperature is varied between 1 eV and 10 eV at atmospheric pressure ($p = 101.3\,\text{kPa}$), while keeping the other parameters fixed. Here, all values are shifted upwards for rising electron temperatures. Simultaneously, the overall shape within each energy block remains constant. Slight variations between the energy blocks exist, but in general the electron temperature acts as a vertical offset

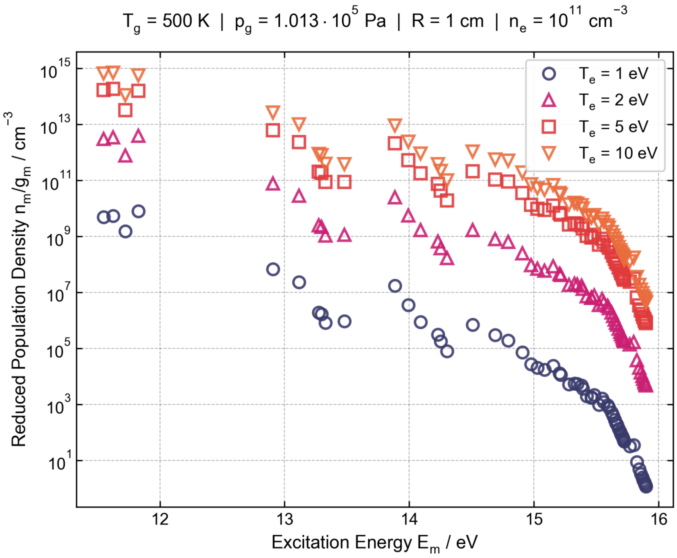

Figure 2.3: Effect of varying electron temperatures on the reduced population densities. For this regime, a change in electron temperature causes a vertical offset and only slight variations in the shape of the curve.

of the graph within the given regime. Importantly, this is only true for the given set of parameters. For other regimes, other dependencies exist and will be accommodated by the CRM.

Lastly, the gas temperature is varied between 500 K and 5000 K, while keeping the other parameters fixed at an electron temperature of 5 eV. Similar to the variation of the discharge pressure, two trends coexist. The overall reduced population density curve experiences a vertical downwards shift for higher gas temperatures. This offset appears much smaller than for changes of the other two parameters on this logarithmic plot but still causes a significant reduction. It is caused by the ideal gas law that is employed to calculate the ground state density from the discharge pressure and gas temperature. Higher gas temperatures result in smaller ground state densities, which lowers the upward flux into the excited levels. In addition to this vertical shift, the linear slope within the energy blocks also varies with the temperature. For higher gas temperatures, the Maxwell-Boltzmann velocity distribution moves towards higher energies, causing fewer atomic de-excitations. Accordingly, the linear slope for the states within one energy block decreases in magnitude.

Having demonstrated that the model accurately simulates the expected physical trends, the results of this implementation are compared to those of Akatsuka [7]. As the authors also employed the approach of Vlcek [6], the two models are expected to produce identical results.

First, the plots for two different discharge pressures are compared to the data set from the literature (see Fig. 2.5). For the 4p block and the 3d/5s block, the two

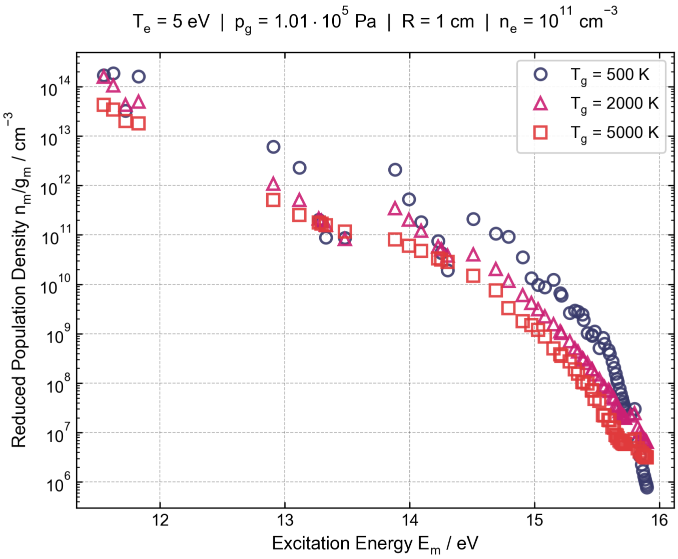

Figure 2.4: Reduced population densities as a function of gas temperature. For higher gas temperatures the whole curve moves downwards. At the same time, the linear slope within the separate energy blocks becomes shallower.

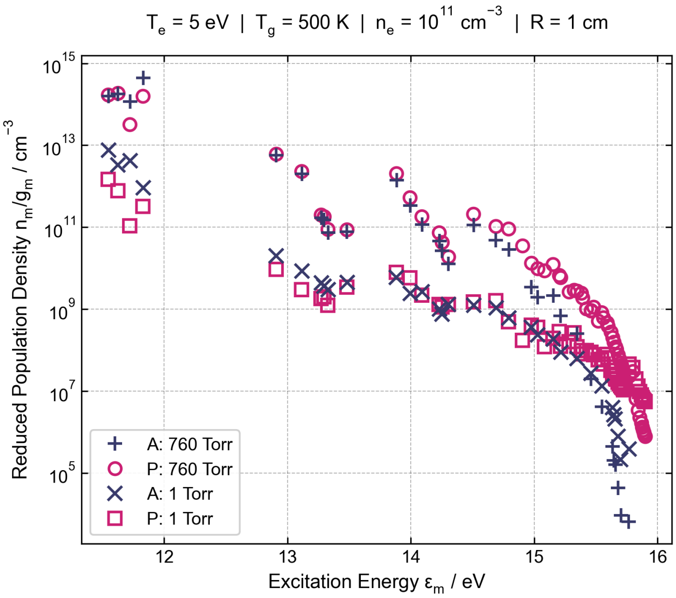

Figure 2.5: The simulation results are compared with the data by Akatsuka [7] for high- and low-pressure plasmas. While the agreement for the atmospheric plasma is a slightly better, both plots show similar deviations for low and high excitation energies. The 4s states have lower population densities than expected, while the highly excited states are significantly overestimated.

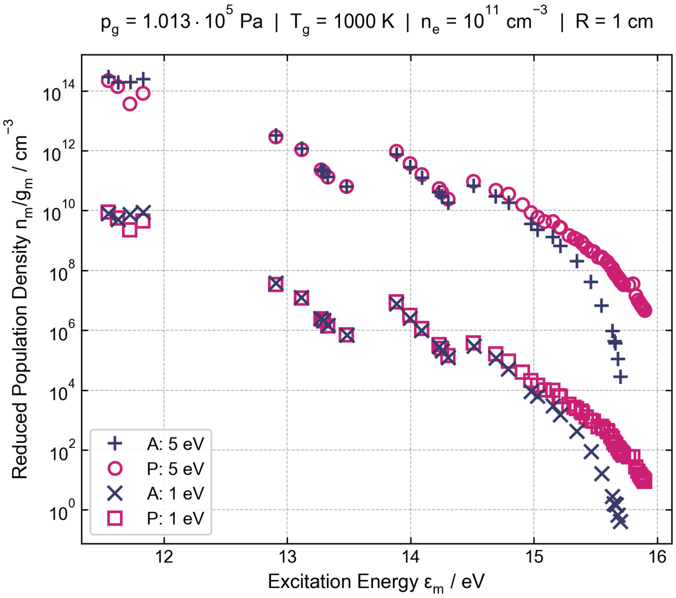

Figure 2.6: At fixed atmospheric pressure, the data is compared for two different electron temperatures. Again, the higher excitation states are overestimated by the CRM, while the rest of the data shows good agreement.

models agree well with each other. There is a slight negative offset towards lower and a positive offset towards higher excitation energies, but the overall shape and magnitude show good agreement. Notably, the two offsets are not caused by discrepancies for the transitions between the given blocks, but rather originate from adjacent energy blocks. Especially for the low-pressure case, the 4s states are one order of magnitude higher for the comparison data. This is most probably caused by a different handling of the optical escape factor for the transitions to the ground state. Here, the implementation followed Bogaerts et al. [8], while Akatsuka does not specify their exact methodology. Furthermore, a large deviation from the literature data occurs for the highly excited states, above approximately 15 eV. For these states, the model largely overestimates the reduced number density by several orders of magnitude. The reason for this is investigated further at the end of this section, by using population rate data provided by the literature source.

Second, the pressure is fixed at one atmosphere and the data is compared for two different electron temperatures (see Fig. 2.6). This data is especially important, since the electron temperature parameter is optimized later on. Parameter-dependent discrepancies would obscure the underlying physical dynamics of the system. Similarly to above, the alignment with the literature data is best for the 4p and 3d/5s energy blocks. The doubled gas temperature seems to bring the two data sets even closer together, so that the 4s states also align very well. Only for the second metastable state is the number density considerably lower than in the literature. There could be an incorrectly implemented or absent transition for this state in particular. As expected, the number density overestimation is again present for the high excitation energies.

To understand the visually detected deviations better, the depopulation frequencies and population rates published by Akatsuka [7] for the levels 6, 13, and 53 are compared to the ones produced by the newly implemented model. For this task, the corresponding values are contrasted in Tab. 2.2. The depopulation mechanisms are well-suited for comparison because they are independent of any population densities. Conversely, the population processes are inherently dependent on the population densities of the respective outgoing states. The comparison shows that the frequencies for atomic excitation and de-excitation align almost perfectly. For the processes by electron collision, the discrepancy between the two data sets is substantially larger. Especially considering de-excitations from level 6 and 13, the frequencies are up to 4.5 times smaller than the values from literature. This suggests that fewer paths for electronic de-excitation exist than in the comparative model. Lastly, the transition probabilities to lower levels are very similar for level 6 (for which reliable NIST data is available) and are considerably lower for the other two levels. While Akatsuka only mentions using NIST data for the transition probabilities, this implementation also included those computed with J_c-K-coupling. Still, Akatsuka reports higher values, suggesting that they used additional undisclosed databases. For the population rates from other levels, the levels 6 and 13 show an agreement that is better than expected considering the discrepancies seen in the depopulation frequencies. Therefore, the overall dynamics balance out to form two highly comparable systems. However, the comparison for level 53 again signifies the heavy overpopulation of the higher excited states. The surrounding states all have much higher population densities than the literature data, which causes these high rates. Nevertheless, the depopulation frequencies themselves do not suggest a highly overestimated upward flux in the system. Instead, the reason for this large positive offset must lie outside the state-to-state interactions.

Based on the visual BP comparison and the discussion of the depopulation frequencies, the reason for the elevated population densities can be narrowed down to two main possibilities. First, there is no comparison data for the continuum interactions, i.e., all ionization and recombination terms. The Rydberg states lie so close to the continuum that ionization processes are expected to occur with high probability. If this is underestimated, the atoms could pile up in these states. Although considerable care was taken in implementing the CRM, it is still possible that one of the equations is not computed properly, or that incorrect assumptions were introduced. Second, the implemented model stops at level 65 with $n = 19$, disregarding all higher excited states that could also be excited and deliver excited atoms even closer to the continuum. Since no source for a boundary condition was found in the literature, no additional ionization and recombination terms were added to the uppermost states in this implementation. This could also act as an upward sink for the accumulating high energy states.

In conclusion, the comparative analysis demonstrates that the newly implemented CRM captures the underlying physical reality with high accuracy. Given the complexity of the system, which consist of 14 distinct transition mechanisms across 65 energy levels, the overall agreement with established literature is remarkable. Although the deviations at higher excitation energies present opportunities for future research, the core state-to-state dynamics are very accurate. Additionally, the open-source nature and clear architecture of the code guarantee transparency and enable future expansion of the CRM. Having validated the model's physical accuracy, the following chapter demonstrates its practical application in deriving the electron temperature from experimental OES measurements.

Table 2.2: Comparison of depopulation frequencies and population rates between Akatsuka [7] and this work. The depicted values are taken from a simulation performed at $T_e = 5\,\mathrm{eV}$, $n_e = 1 \cdot 10^{11}\,\mathrm{cm}^{-2}$, $T_g = 500\,\mathrm{K}$ and $p_g = 1 \cdot 10^5\,\mathrm{Pa}$. Read all exponential terms like $9.6\mathrm{E}5 = 9.6 \cdot 10^5$.

	Level 6		**Level 13**		**Level 53**	
Mechanism	Akatsuka	This Work	Akatsuka	This Work	Akatsuka	This Work
Depopulation Frequency ($\mathbf{s^{-1}}$)						
Electron Excitation	9.6E5	1.48E6	3.4E5	3.15E5	3.1E8	3.94E8
Atom Excitation	7.5E5	7.52E5	3.7E7	3.69E7	2.7E10	2.80E10
Spontaneous Emission	2.3E7	2.55E7	1.6E7	4.81E6	1.1E5	5.72E4
Electron De-Excitation	4.7E4	1.96E4	1.5E5	3.33E4	1.8E8	2.25E8
Atom De-Excitation	3.0E7	3.03E7	1.8E8	1.84E8	1.6E10	1.58E10
Population Rate ($\mathbf{cm^{-3}s^{-1}}$)						
Spontaneous Emission	1.2E20	1.23E20	5.9E18	8.46E18	5.4E11	2.74E15
Electron De-Excitation	1.1E18	6.34E17	2.4E17	3.83E17	8.3E15	3.16E19
Atom De-Excitation	6.8E20	8.05E20	8.0E20	1.29E21	7.9E17	2.98E21
Electron Excitation	1.4E20	1.31E20	2.3E20	2.22E20	4.3E16	3.28E19
Atom Excitation	1.6E11	5.90E10	2.7E20	3.89E20	1.8E18	1.60E21

3 Experimental Methodology and Results

This chapter is dedicated to the experimental part of the project, where the CRM is applied to a real setup for determination of the electron temperature. The setup is designed and operated by one of the PhD students in the laboratory at the KIT. First, the geometrical and electrical setup for the plasma formation are explained. Next, the spectral analysis is outlined and separated into the hardware and software part. In the last section the experimental spectral analysis will be combined with the theoretical CRM to retrieve the electron temperature.

3.1 Plasma Apparatus

A significant advantage of APPs is that they can be created in the open atmosphere and do not require any expensive vacuum infrastructure. The plasma vessel used in this project is a glass test tube, which is open at the bottom and has a separate T-junction for the gas inlet (see Fig. 3.1). Moreover, the top of the test tube is sealed with a rubber cap that has a hole in the center to hold a thin straight wire in place. Towards the bottom, the tube tapers to a smaller diameter. Here, a thin copper sheet is firmly wrapped around the tube, forming the outer electrode. This geometry forms a cylindrical DBD configuration, where the central wire is the inner electrode and the borosilicate glass acts as the dielectric barrier (see Fig. 3.2).

The argon gas is supplied by the laboratory gas supply, which is pressurized to 0.2 MPa. Before the gas is routed into the discharge chamber, a rotameter is used to control the flow rate, which is usually set to $0.7\,\mathrm{L\,min^{-1}}$. It flows axially through the configuration, exiting partially ionized at the bottom of the tube. This is where a water sample containing phytoplankton is placed for the main experiment.

Electrically, the electrodes are supplied by an AC voltage power supply typically operating at an amplitude of around 2 kV and a frequency of 10 kHz to 20 kHz. The electrical connection is established using a laboratory cable fitted with an alligator clip. This allows for quick changes to the experimental setup, since several other DBD configurations are used.

For optical access to the plasma, a small rectangular aperture of 1 mm times 2 mm is cut into the axial mid-point of the outer electrode. This allows for a measurement of the optical properties of the bulk plasma, which is expected to be representative of the entire discharge. Due to the small dimensions of the optical window compared to the length and circumference of the outer electrode, the impact on the field structure is assumed to be negligible.

During operation, two main problems with the described experimental setup were identified. First, the inner electrode is only securely centered at the uppermost point of the glass tube. This makes it impossible to perfectly align the electrode along the central axis. In the future, alignment spacers should be implemented at several points within the tube to make the setup reproducible. A challenge will be avoiding disturbances to and ensuring that the alignment spacers can withstand the conditions within the tube. Second, the open system allowed atmospheric gases such as nitrogen and water vapor to enter the discharge chamber. This caused significant contamination of the optical spectrum, obscuring some relevant argon emission lines, as will be discussed in section 3.2. Enclosing the experimental setup in a sealed chamber is therefore recommended, so the OES measurement can be performed under a pure argon atmosphere.

3.2 Spectral Analysis

For the spectral analysis, the PMA-12 PHOTONIC MULTICHANNEL ANALYZER from Hamamatsu is used. It combines a Czerny-Turner spectrometer with a Back-Thinned Charged-Coupled Device (BT-CCD) linear image sensor featuring 1025 channels. The BT-CCD sensor offers the advantages of high quantum efficiency, smooth spectral response, and a high signal-to-noise ratio [24]. A 1.5 m long bundled optical fiber with a diameter of 1 mm feeds the light into the spectrometer. The optical probe at the end of the fiber lacks collection optics and houses only the bare end of the fiber. Because of the high sensitivity, it is sufficient to bring this probe close to the sample to acquire a spectrum. Since the probe is used without collection optics, all spectra represent the spatially integrated emission from the field of view of the optical fiber.

In operation, the spectrometer is connected to a laptop via USB-A and controlled with the associated PMA SOFTWARE from Hamamatsu. This software automatically applies all spectral response corrections and performs a dark current correction for every exposure. The exposure time can be varied between 19 ms and 64 s, with a saturation limit of 65535 counts. Within the wavelength range between 200 nm and 950 nm, the spectrometer has an accuracy of ± 0.75 nm and a resolution better than 2 nm [25].

A typical emission spectrum of the APP setup is shown in Fig. 3.3. The spectrum is displayed on a semi-logarithmic plot to enhance the visibility of small peaks and visualize the changing baseline. Within the range of 650 nm to 930 nm, the argon emission lines from the 4p→4s system are clearly visible. However, the 5p→4s emission lines, which would be expected within the range of 390 nm to 480 nm, are obscured by parasitic emissions. These originate from the second positive system of nitrogen, which describes the vibronic transitions $C^3\Pi_u \rightarrow B^3\Pi_g$ [26]. Addi-

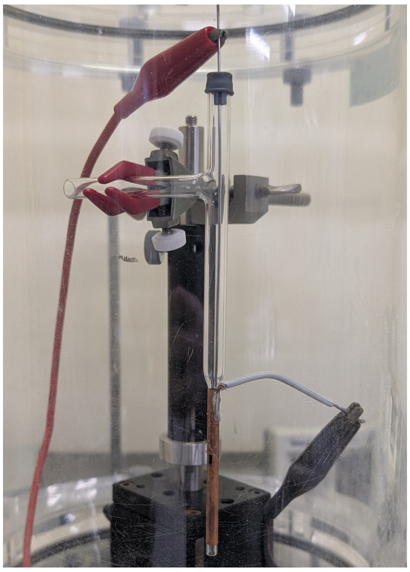

Figure 3.1: Photograph of the glass tube used for generating the APP. Here, it is placed inside a sealed chamber, which is recommended for better control of the gas environment. However, the chamber is not required for operating the system.

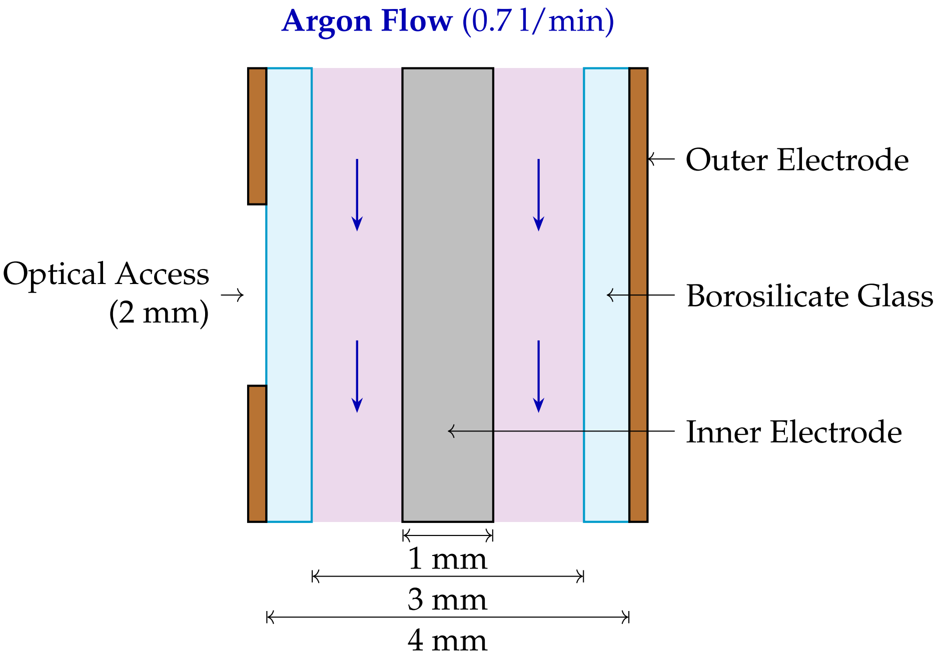

Figure 3.2: This schematic presents the experimental setup. Argon gas flows axially through a cylindrical DBD geometry, where the gas is ionized and a plasma is formed. All geometrical parameters are drawn to scale, except for the channel length. The actual electrodes have a length of 6 cm.

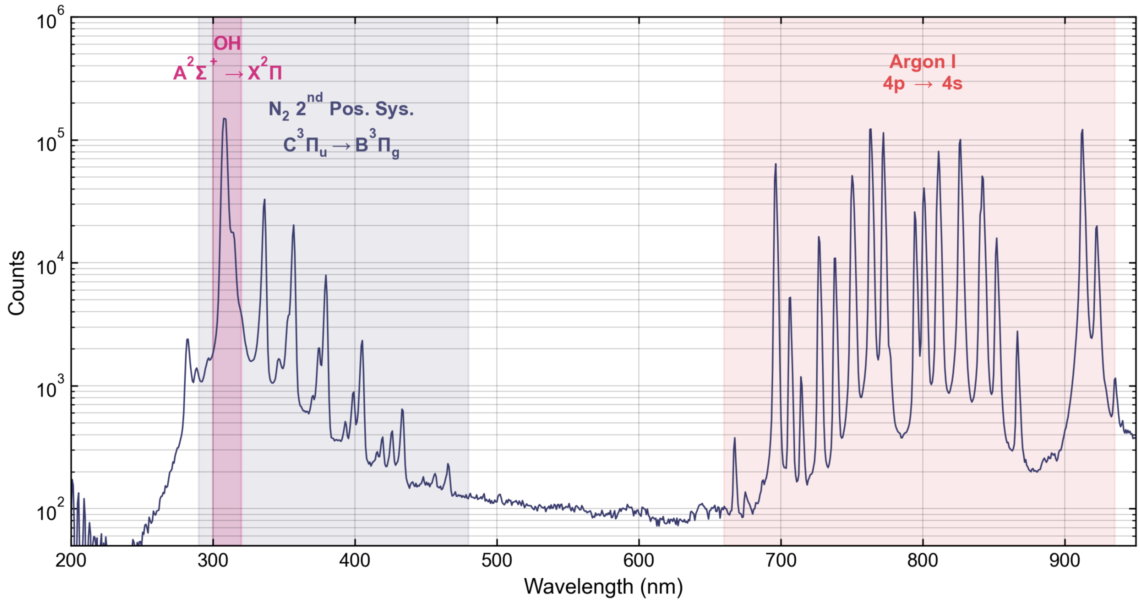

Figure 3.3: This is a typical emission spectrum for the APP used in this project. The semi-log plot allows for simultaneous visualization of the strongest and weakest peaks in the spectrum. The three main visible transitions are the electronic argon emission lines and the vibronic OH and N_2 emission systems.

tionally, the characteristic OH transition $A^2\Sigma^+ \rightarrow X^2\Pi$ appears as the most intense peak in the spectrum around 308 nm. From the washed-out shape of the peaks, the low spectral resolution of the spectrometer is evident.

Due to the strong parasitic molecular transitions, only the strongest argon lines originating from the fine-structure 4p state are visible. However, to improve the accuracy of the optimization that will be explained in section 3.3, it would be advantageous to have access to line data for excited states outside the 4p block. Therefore, it is suggested to perform the OES measurements under a pure argon atmosphere. Preliminary tests were performed with a hard-plastic chamber, but the sealing and argon supply require improvement to enable reliable measurements. To demonstrate the advantages of a pure argon emission spectrum in the following optimization procedure, an optical emission spectrum was taken from another experiment that was simultaneously performed in the laboratory. This experiment featured an inductively coupled low-pressure argon plasma at around 3 Pa. Here, the 5p→4s and several other transitions are visible (see Fig. 3.4). This is, in part, due to the higher electron temperature of the LPP, but also due to the absence of molecular emissions. The next section will outline how this data is analyzed to automatically relate the emission lines to the lumped levels and calculate the respective relative population densities.

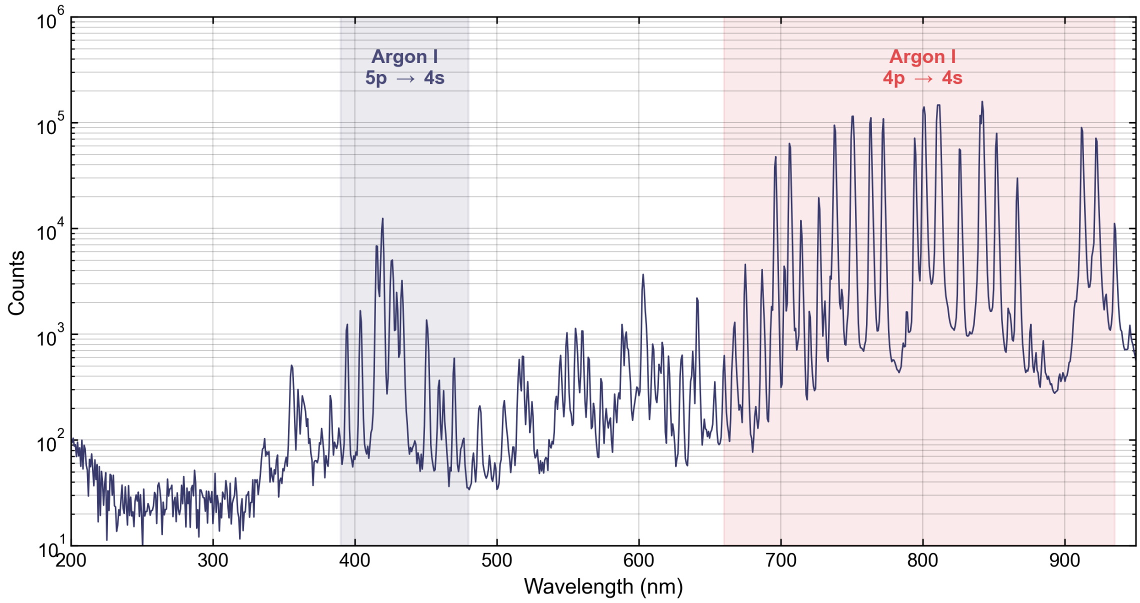

Figure 3.4: In contrast to the APP, no parasitic molecular transitions are visible. Instead, many emission lines belonging to several higher excited states outside the 4p block are resolved.

3.3 Application of the Collisional-Radiative Model

For the spectral analysis, a custom PYTHON package was developed to calculate the relative population densities for all clear spectral lines visible in any argon emission spectrum[1]. Similar to the CRM code, the user interface is provided by a config file that specifies all relevant parameters and is loaded through the terminal. The code will automatically switch between a manual and an automatic mode.

In the manual mode, the user must input all wavelengths for which the peaks should be processed. The program will then iterate through all wavelengths, search for peaks that lie within a certain window around that wavelength and integrate the total counts of every peak. An integration window is centered around the detected peak with a total width determined by the Full Width Half Maximum (FWHM) multiplied by a custom factor. The integration is then performed by summing the total area under the curve numerically using the trapezoidal rule, while subtracting a linear baseline that is drawn by connecting the two outermost points. Next, the calculated line intensity is related to the number density of the given fine-structure upper state by dividing by the corresponding transition probability. For both the state configuration and the transition probability, the desired wavelength is related to the corresponding line specified in the NIST atomic spectral database [15]. Lastly, the fine-structure upper state configuration is mapped to

1 The spectral analysis code is published on https://github.com/peterplr/spec-pop.

Vlcek's lumped energy levels. This involves a two-step process: First, all number densities that correspond to the exact same state are averaged to one value. Second, all fine-structure densities that belong to the same lumped level are summed up. If some lines belonging to a level are missing or cannot be processed, the summed number density is scaled up to the full shell using the statistical weights. This assumes the remaining states within the level to be in perfect thermodynamic equilibrium with the others, which is not always a good assumption. Nevertheless, it is the only way to approximate the number density of lumped levels that are not completely defined by separate spectral lines.

Conversely, the automatic mode is triggered when no wavelengths are specified manually in the config file. In this case, the algorithm iterates over all peaks that are found in the spectrum for the custom wavelength range and minimum peak height specified in the config file. During peak detection, the peak must be mapped to the correct NIST line. The algorithm checks whether a single peak or multiple peaks exist within the given range. If there are several NIST lines specified in the range, the code checks whether the relative intensity of the dominant line and all other lines in the window differ by a factor of 100 or more. Every peak that does not pass this condition is categorized as ambiguous. This is particularly important since the low-resolution spectrum will merge any close-lying lines of comparable intensity, making a distribution of the intensity among the respective upper levels impossible.

There are two plotting options that can be activated to monitor the algorithm and iteratively find the best options for integration and peak detection. First, every individual line integration is visualized in a separate plot so that the integration window and proper peak detection can be checked. Second, a large overview plot is created in which all integration areas are marked. This helps to identify overlaps between integration areas. Both plots can be interactively enabled or disabled in the config. However, it is recommended to always turn on the plot overview, so any mistakes in the processing are detected instantly.

For a first independent analysis of the spectrum for the APP, the automatic mode is used in the range between 650 nm and 940 nm. A total of fourteen lines are detected, and the respective overview plot is depicted in Fig. 3.5. The corresponding population density data is tabulated in Tab. 3.1. The overview shows that the integration windows contain the main structure of the peaks very well and that several peaks were correctly excluded due to ambiguity concerns. Moreover, the tabulated values give the relative densities and the averaging results for lines originating from the same fine-structure state. Comparing these averaged lines, one can observe a drop-off towards infrared wavelengths. For instance, the two >900 nm lines for level 7 show a reduction by 30 % and 26 % compared to the corresponding shorter wavelengths. However, since the lines originate from exactly the same state, the relative densities should be almost identical, deviating only by the measurement and transition probability error. This suggests that the response calibration for the spectrometer is no longer accurately correcting the offset. Con-

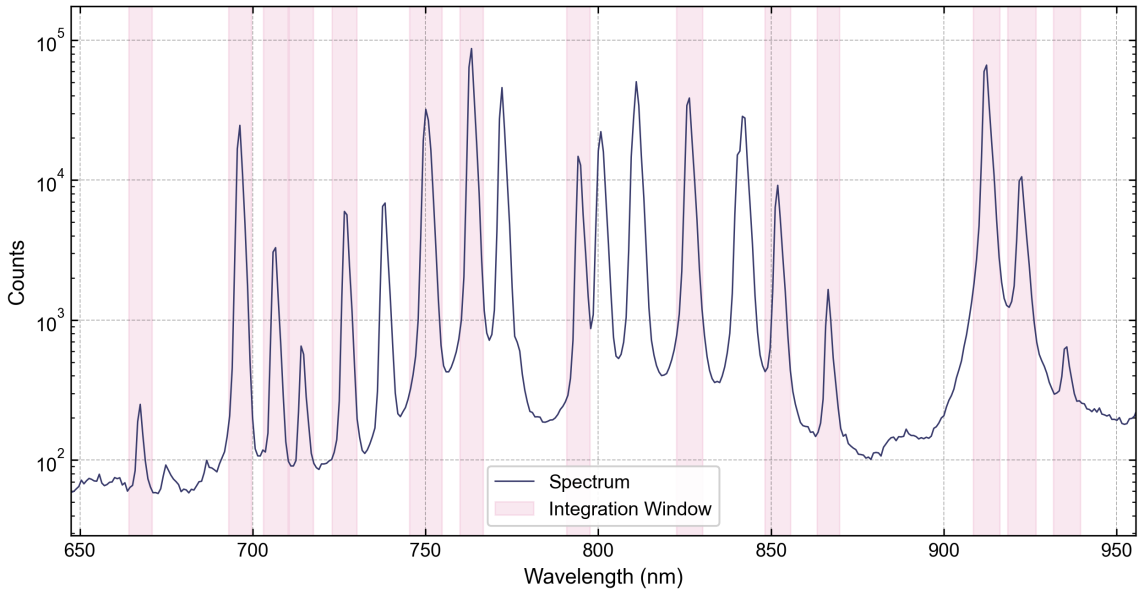

Figure 3.5: Emission spectrum of the APP viewed through the window cutout. The shaded areas mark the integration windows for all non-ambiguous lines.

Table 3.1: Relative population densities and their averaged values for the APP grouped by the level designation n. All lines that originate from the same upper state are averaged, and their relative standard deviation is given.

n	Wavelength in nm	Relative Density in a.u.	Average Density in a.u.	Relative Error in %
6	912.55	7.56	–	–
7	763.72	6.77	5.77	25
	922.70	4.76	5.77	25
	867.03	1.07	0.928	21
	935.68	0.792	0.928	21
8	706.92	1.73	–	–
	714.90	1.69	1.50	13
	795.04	1.49	1.50	13
	852.38	1.31	1.50	13
9	696.74	6.95	6.32	12
	727.49	6.58	6.32	12
	826.68	5.44	6.32	12
11	667.91	1.59	1.67	6
	750.59	1.74	1.67	6

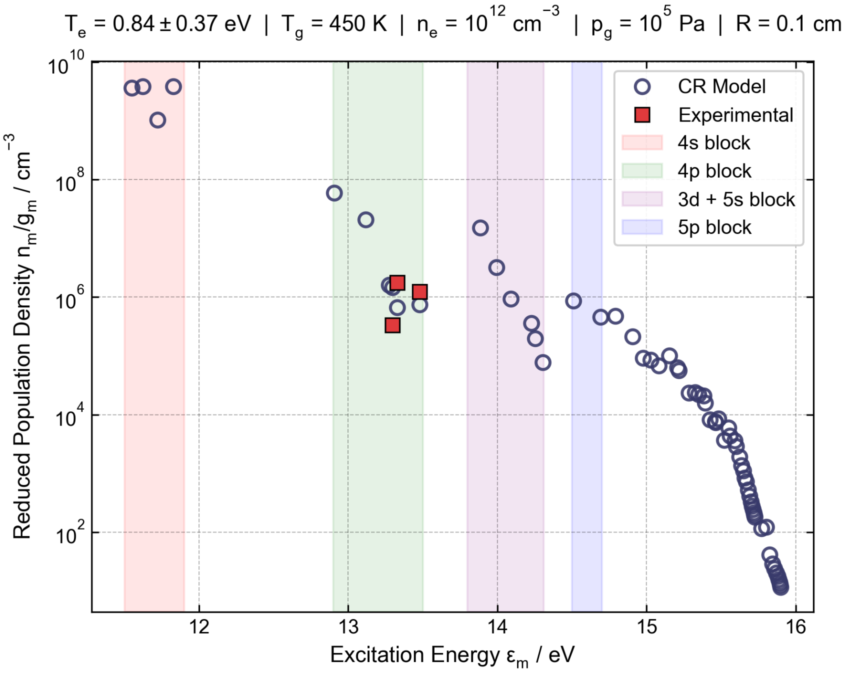

Figure 3.6: BP of the optimal electron temperature when compared to the experimental data. The experimental values are only relative, so they are visualized by performing a fit to the theoretical curve.

sidering that the spectrometer has not been recalibrated since its purchase in 2015, this discrepancy is expected. Therefore, it is strongly recommended that the spectrometer be recalibrated. In the meantime, all lines with wavelengths longer than 730 nm are excluded from the calculations.

For calculating the electron temperature of the APP, the relative number densities are provided to the optimizer of the CRM. Levels 8, 9, and 11 are experimentally determined to have the aggregate number densities 3.42, 6.77, and 1.59, respectively. The pressure is set to $1 \cdot 10^5$ Pa, the radius of the plasma vessel to 1 mm, and the gas temperature is approximated as 450 K. To determine the latter value, the molecular emission lines are evaluated using the massiveOES PYTHON package [27]. This code calculates the rotational and vibrational temperatures of the plasma by fitting a theoretical spectrum, similar to the optimization procedure employed in the CRM. Assuming the rotational temperature to be in thermal equilibrium with the gas temperature, the value of 450 K is deduced.

Since the electron density is unknown, three different values ranging from $1 \cdot 10^{10}\,\mathrm{cm}^{-3}$ to $1 \cdot 10^{12}\,\mathrm{cm}^{-3}$ are evaluated during optimization. Correspondingly, the optimizer found electron temperatures of 0.93 ± 0.83 eV, 0.90 ± 0.58 eV, and 0.84 ± 0.37 eV, respectively. The BP for the largest electron density is shown in Fig. 3.6, where the relative experimental densities are fit to the curve. This plot reveals

Table 3.2: Relative aggregate and reduced population densities for the LPP grouped by the level designation n.

n	Aggregate Density in a.u.	Reduced Density in a.u.
8	60.0	7.50
9	15.3	5.10
11	11.8	11.8
18	56.6	2.36
19	20.3	1.69
20	38.5	0.802
21	6.19	0.258
27	19.0	0.396

the reason behind the large errors in the calculated electron temperatures. First, the reduced number density of level 8 is lower than those of levels 9 and 11, when compared to the theoretical expectation. This introduces a large error that the simulation will never be able to minimize because the ratio between the levels will be inverted for all electron temperatures. Second, the three experimentally specified levels lie in an excitation energy window of only $\Delta\varepsilon = 0.185\,\text{eV}$. This gives the optimizer limited leverage, since the three levels will always react very similarly to changes in temperature. Therefore, it becomes clear why a spectrum without molecular impurities and the resulting measurement of higher energy states would be very advantageous for the optimization.

To illustrate the advantage of having experimental number densities over a larger energy window, the spectrum of the above-mentioned LPP is evaluated analogously to that of the APP. Since the same spectrometer is used, the same wavelength cutoff at 730 nm is applied. The calculated relative densities are displayed in Tab. 3.2. For the other parameters, a pressure of 3 Pa was measured, a radius of 8 cm is approximated, and a room temperature of 300 K is assumed. This time, optimizations are performed for three electron densities in the range $1 \cdot 10^{11}\,\text{cm}^{-3}$ to $1 \cdot 10^{13}\,\text{cm}^{-3}$. Correspondingly, the optimizer found electron temperatures of $0.774 \pm 0.023\,\text{eV}$, $2.25 \pm 0.61\,\text{eV}$, and $2.80 \pm 0.73\,\text{eV}$, respectively. Although the optimization for the lowest electron density produces a mathematically precise electron temperature, this solution is nonphysical and is disregarded for the given inductively coupled plasma. While the other two optimizations show larger errors, the magnitude between 2 eV and 3 eV fits very well into the physically expected range. The BP for the electron density of $1 \cdot 10^{12}\,\text{cm}^{-3}$ shows the significant advantage of having experimental data that is spread over a range of about 2 eV (see Fig. 3.7). In contrast to the APP, the experiment and simulation for the LPP agree very well, allowing for high confidence in the optimized temperature.

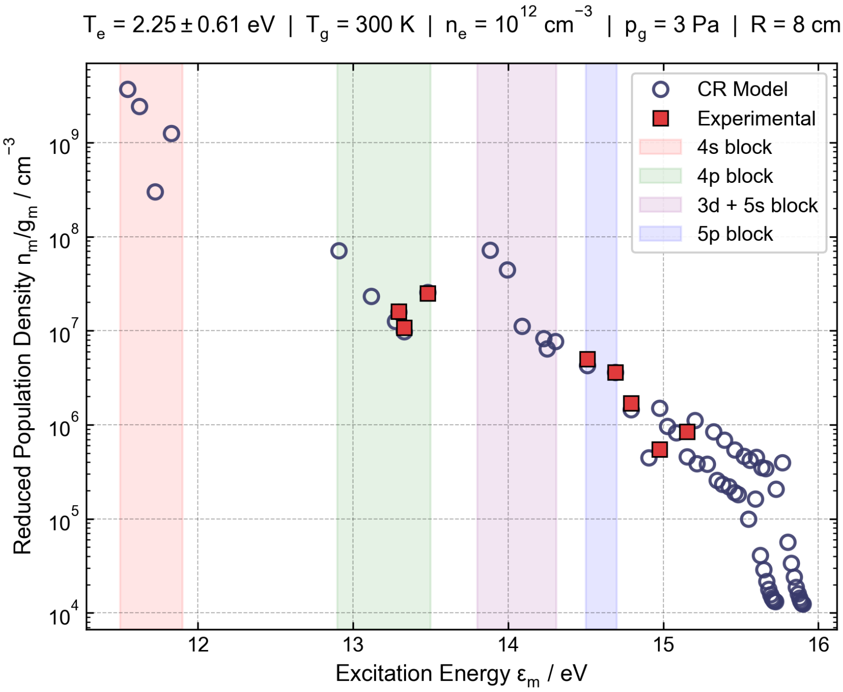

Figure 3.7: For the LPP, the experimental data ranges over excitation energies with a width of about 2 eV. This gives the optimizer sufficient data points to find the electron temperature with higher certainty. The agreement between the experimental and theoretical data is very good.

In conclusion, the optimization for the spectrum of the LPP showed that the general method of fitting the CRM output to the experimental data is highly effective. Meanwhile, the APP optimization failed to produce trustworthy results due to the small number and energy range of experimental density values. However, if the experiment is successfully transferred into a pure argon environment, good results are expected to be obtained. Furthermore, the seamless interchangeability between the low- and high-pressure plasma emphasizes the flexibility of the setup. A single spectrum of any argon plasma is sufficient to calculate the electron temperature.

4 Conclusion

In this project, a highly modular PYTHON package containing a comprehensive CRM for argon was developed. Parameter studies demonstrated that the model correctly captures the complex physical behavior of the atomic system. Furthermore, a comparison with the simulation results from Akatsuka [7] revealed a strong agreement between the two models in the core energy levels. Additionally, discrepancies for the lower and higher energy levels were found and and their root causes were isolated.

Future research should be dedicated to refining the implemented model, building on the existing modular structure and base transitions. The full simulation code is openly accessible via the GitHub repository and can be redistributed under a public license. There are two approaches to further development of the codebase. First, the legacy implementation from Vlcek [6] could be kept and the discrepancies in the continuum interactions could be resolved. Second, modern cross section databases could be used to re-compute the semi-empirical scaling parameters to align the CRM with current physical data standards.

For the spectral analysis, a highly flexible software package was developed to extract population density data from any argon emission spectrum. However, reliability of any spectral measurement fundamentally depends on the hardware. Experimentally, available equipment was limited to a spectrometer with an outdated calibration and no calibration hardware. This restricted the analytical spectral range, artificially lowering the number of data points. Additionally, the APP setup was designed for an application under ambient atmospheric conditions, causing problems in the spectral analysis due to molecular impurities. Nevertheless, after implementing the recommended experimental and calibration changes, the derivation of the electron temperature should yield highly accurate results.

By applying the CRM to two experimental datasets, the general functionality of the optimization algorithm was successfully demonstrated. Although the analysis of the APP yielded inconclusive results due to the spectral limitations mentioned above, the model proved highly effective when applied to the LPP. In the future, two main approaches can further improve the accuracy of the measurement system. First, refining the theoretical CRM could stabilize the population density results and enhance their physical accuracy. Second, improving the spectral analysis should make the spectral data itself more coherent and improve the statistics by increasing the usable wavelength range. Ultimately, the developed framework provides the collaborating research groups with a highly adaptable, open-source foundation for spectroscopic diagnostics of argon plasmas.

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